

# Liver imaging report — FDG PET-CT of 19 June 2026

Anastasia Zherlitsyna, born 25.04.1976 — metastatic GIST with liver metastases

Compiled 10 September 2026 from the patient's records (docs\_eng archive)

## Contents

|                                                                                                                                                                                                  |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1 Purpose and sources</b>                                                                                                                                                                     | <b>2</b>  |
| <b>2 Clinical summary</b>                                                                                                                                                                        | <b>2</b>  |
| <b>3 The 19.06.2026 PET-CT: three readings</b>                                                                                                                                                   | <b>3</b>  |
| <b>4 Independent image analysis of the 19.06.2026 study</b>                                                                                                                                      | <b>4</b>  |
| 4.1 Data and method . . . . .                                                                                                                                                                    | 4         |
| 4.2 Findings . . . . .                                                                                                                                                                           | 4         |
| 4.3 3D reconstruction . . . . .                                                                                                                                                                  | 7         |
| <b>5 Evolution of the liver disease 2023–2026</b>                                                                                                                                                | <b>8</b>  |
| <b>6 Gastric-wall GIST, uterus and right ovarian endometrioma 2022–2026</b>                                                                                                                      | <b>9</b>  |
| 6.1 Gastric-wall mass (primary GIST site) . . . . .                                                                                                                                              | 9         |
| 6.2 Uterus with leiomyomas and right adnexal mass (whole pelvic complex traced) . . . . .                                                                                                        | 10        |
| <b>7 Feasibility of liver-directed treatment</b>                                                                                                                                                 | <b>12</b> |
| 7.1 Opinion of the interventional radiologist (Dr. Oleksandr Korshak, Shalimov National Scientific Centre of Surgery and Transplantology, Kyiv; 10–11.09.2026, messenger consultation) . . . . . | 12        |
| 7.2 Guideline basis for local treatment of focal progression . . . . .                                                                                                                           | 12        |
| 7.3 Evidence summary (OpenEvidence review, 05.09.2026) . . . . .                                                                                                                                 | 12        |
| 7.4 Proposed systemic drugs and coverage of the three KIT clones . . . . .                                                                                                                       | 13        |
| <b>8 Questions for the multidisciplinary tumour board</b>                                                                                                                                        | <b>13</b> |
| <b>9 Appendix A — Expert opinion on the FDG PET-CT, 08.09.2026 (English translation)</b>                                                                                                         | <b>15</b> |
| <b>10 Appendix B — LISOD retrospective radiology report, 08.07.2026 (English translation)</b>                                                                                                    | <b>17</b> |
| <b>11 Appendix C — PET-CT report, Feofaniya, 19.06.2026 (English translation)</b>                                                                                                                | <b>18</b> |
| <b>12 Appendix D — Slice-by-slice verification of the lesion outline</b>                                                                                                                         | <b>20</b> |

# 1 Purpose and sources

This report gathers, in one place, everything that the available imaging says about the liver disease of the patient as of September 2026, so that an interventional radiologist, hepatobiliary surgeon or oncologist can judge the feasibility of liver-directed treatment without opening the full archive. It contains: (1) the three professional readings of the 19.06.2026 FDG PET-CT; (2) an independent quantitative image analysis of that study with raw and labelled CT slices and a 3D reconstruction; (3) the evolution of the liver lesions across the five PET-CTs of 2023–2026; (4) the opinion of the thermal-ablation specialist consulted on 10.09.2026; (5) a summary of the treatment-option evidence reviewed with OpenEvidence on 05.09.2026.

The image analysis (sections 4–7) was produced by the patient’s family with open-source software on the original DICOM data and is **not** a radiologist’s report. It is included to make the anatomy explicit; measurements should be confirmed on fresh imaging before any procedure.

Sources are held on the family’s Google Drive (`docs_eng/`): the Ukrainian originals, English translations, DICOM discs (`anastasia_disks/`) and the overview document `Overview.docx`.

Direct links (shared Google Drive): [analysis folder](#) · [rotation animation](#) · [interactive 3D model](#) · [DICOM disc of 19.06.2026](#) · [Overview.docx](#) · [OpenEvidence review PDF](#) · [FoundationOne report](#).

# 2 Clinical summary

**Formulated diagnosis.** ICD-10 C16.9; ICD-O 8936/3. Gastrointestinal stromal tumour of the stomach, KIT exon 11 deletion (Q556\_V559>H), with bilobar liver and peritoneal metastases, stage IV; initially (08/2022) treated as leiomyosarcoma of the uterine body T2NxM1, revised 04–06/2023 on histology. Status after gemcitabine/docetaxel/bevacizumab ×4 (progression), pazopanib 05–12/01/2023 (PRES with left occipital haemorrhage), trabectedin ×3 (progression, PET-CT 11/04/2023); imatinib 400 mg/day from 03/05/2023 with marked response (08/2023) and stable disease (03/2024, 05/2025); 06–09/2026 return of FDG-active mural components in the liver lesions (metabolic progression per the 08/09/2026 expert reading); FoundationOne Liquid CDx 08/2026: secondary KIT exon 13 V654A and exon 18 A829P on top of the exon 11 driver (ctDNA fraction 32 %); sunitinib recommended 08/09/2026, not yet started. Concomitant: uterine leiomyomas, right ovarian endometrioma, type 2 diabetes, iron-deficiency anaemia.

**Patient.** Anastasia Zherlitsyna, born 25.04.1976, Kyiv, Ukraine.

**Diagnosis.** Metastatic GIST of the gastric wall with liver metastases; KIT exon 11 deletion (Q556\_V559>H). First treated (from 28.09.2022) as uterine leiomyosarcoma T2NxM1; GIST was established by the fourth biopsy (21–26.04.2023) and confirmed by pathology revision in Israel (04.06.2023). Uterine masses are considered leiomyoma; right ovarian endometrioma.

**Pathology.** Liver metastasis biopsies of 29.08.2022 and 24.01.2023 were read as leiomyosarcoma (no IHC possible on the first); the 21.04.2023 EUS-FNA/trephine biopsy of the paragastric mass and the 25.04.2023 liver core biopsy showed GIST (DOG-1, CD117 positive, desmin negative); IHC on the original 2022 slide (04.05.2023) and the Israeli revision (04.06.2023: DOG1, CD34, c-kit positive, SDH-B deficit) confirmed GIST. The biopsied liver lobe/segment is not stated in any report.

**Relevant history.** PRES with intracranial haemorrhage on pazopanib (January 2023); port-a-cath; SVC probably occluded with chest collaterals (08.09.2026 reading); normotensive.

| Period              | Systemic treatment                          | Examination / event                                                                                 | Finding                                                                                                        |
|---------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Aug 2022            | —                                           | Liver-metastasis biopsy 29.08.2022; tumour board 31.08.2022                                         | Uterine leiomyosarcoma T2NxM1 (revised to GIST in May 2023)                                                    |
| Sep 2022 – Jan 2023 | Gemcitabine + Docetaxel + Avastin, 4 cycles | MRI 14.10.2022, CT 01.12.2022, CT 04.01.2023                                                        | Progression in the liver                                                                                       |
| Dec 2022 – Jan 2023 | (same)                                      | <a href="#">Oncomine</a> on the 2022 biopsy; <a href="#">BRCA panel</a> ; <a href="#">BAM files</a> | KIT exon 11 p.Gln556_Val559delinsHis (VAF 42.5 %); subclonal KIF5B-MET fusion; TMB low, MSS; germline negative |

| Period         | Systemic treatment                                                              | Examination / event                            | Finding                                                                                                         |
|----------------|---------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 5–12 Jan 2023  | Pazopanib 800 mg (stopped day 8)                                                | Brain MRI 13.01 and 20.01.2023                 | PRES with left occipital haemorrhage                                                                            |
| Feb – Mar 2023 | Trabectedin, 3 cycles                                                           | PET-CT 11.04.2023; biopsies 21–25.04.2023      | Progression; GIST established                                                                                   |
| May – Aug 2023 | Imatinib 400 mg/day (start 03.05.2023)                                          | PET-CT 17.08.2023                              | Pronounced response, loss of activity and solid rim components                                                  |
| Mar 2024       | Imatinib 400 mg/day (cont.)                                                     | PET-CT 14.03.2024, Israeli revision 25.03.2024 | Stable, minimal uptake                                                                                          |
| May 2025       | Imatinib 400 mg/day (cont.)                                                     | PET-CT 13.05.2025, LISOD read 19.05.2025       | PET negative; new inactive 11 mm mural nodule                                                                   |
| 19 Jun 2026    | Imatinib 400 mg/day (cont.)                                                     | PET-CT 19.06.2026                              | First read as stable                                                                                            |
| 8 Jul 2026     | Imatinib 400 mg/day (cont.)                                                     | LISOD retrospective read                       | New FDG-active mural components — progression                                                                   |
| 30 Jul 2026    | Imatinib 400 mg/day (cont.); sunitinib 50 mg 4/2 prescribed, <b>not started</b> | LISOD oncologist; echocardiogram               | Progression documented; EF 67 %                                                                                 |
| 20 Aug 2026    | Imatinib 400 mg/day (cont.)                                                     | FoundationOne Liquid CDx, blood                | ctDNA 32 %; KIT exon 11 (17 %) + exon 13 V654A (8.8 %) + exon 18 A829P (4.0 %); BRCA2, CDKN2A, MTOR; TMB 3, MSS |
| 8 Sep 2026     | Imatinib 400 mg/day (cont.)                                                     | Expert PET-CT reading, Israel                  | Metabolic progression, especially the left-lobe lesion                                                          |
| 8 Sep 2026     | Imatinib 400 mg/day (cont.)                                                     | LISOD oncologist (Ukrainian original)          | Sunitinib 50 mg 4/2 recommended; regorafenib 160 mg d1–21 q28 as an option; neither started as of 11.09.2026    |

### 3 The 19.06.2026 PET-CT: three readings

|                  | Feofaniya, 19.06.2026 (primary read) | LISOD, 08.07.2026 (retrospective read, Dr. M. Novikov)                           | Israel, 08.09.2026 (expert opinion, Dr. V. Kulikova)                          |
|------------------|--------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Documents        | translation (Ukrainian original)     | translation (Ukrainian original)                                                 | translation (Russian original)                                                |
| Comparison       | prior studies                        | 13.05.2025 and 14.03.2024                                                        | 13.05.2025                                                                    |
| Right-lobe mass  | known cystic residue                 | new enhancing, moderately FDG-active mural components; liver enlarged to 18.7 cm | about 15 × 10 cm (was 11–12.5 cm), peripheral uptake somewhat more pronounced |
| Left-lobe lesion | —                                    | mural components with uptake (SUVmax about 3.2 reported)                         | about 42 × 36 mm, size unchanged, <b>uptake intensity increased</b>           |

|                     | Feofaniya, 19.06.2026<br>(primary read) | LISOD, 08.07.2026<br>(retrospective read,<br>Dr. M. Novikov)    | Israel, 08.09.2026<br>(expert opinion, Dr. V.<br>Kulikova)                                                                                                                      |
|---------------------|-----------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gastric-wall mass   | unchanged                               | unchanged,<br>calcifications                                    | about 37 × 31 mm,<br>minimal uptake,<br>unchanged                                                                                                                               |
| Portal vein / ducts | —                                       | —                                                               | portal vein patent;<br>intrahepatic ducts not<br>dilated                                                                                                                        |
| Bones               | —                                       | 31 mm lucent lesion,<br>right femoral head (to<br>characterise) | no pathological foci or<br>destruction in the<br>scanned area                                                                                                                   |
| Conclusion          | initially read as stable                | return of viable mural<br>tumour components —<br>progression    | <b>metabolic progression<br/>of the liver lesions,<br/>especially the left-lobe<br/>lesion; enlargement of<br/>the right-lobe mass<br/>with somewhat<br/>increased activity</b> |

Full texts of the LISOD and Israeli readings are reproduced in the appendices.

## 4 Independent image analysis of the 19.06.2026 study

### 4.1 Data and method

The study contains a low-dose CT, an arterial-phase and a **venous-phase contrast CT (series 9, 2 mm, 412 slices)** and the whole-body FDG PET (SUVbw, 2.67 × 2.67 × 5 mm). The venous-phase CT and the PET were converted from DICOM with the correct patient geometry and fused in world coordinates. Organ, vessel and tumour masks were produced with TotalSegmentator on the corrected volume and reviewed slice by slice; the lesion outline is a density threshold (hypodense relative to enhanced liver) seeded at the FDG peak, and the FDG-active volume is the connected region above 50 % of the peak-to-background contrast. Display convention: radiological (anterior up, patient left on image right).

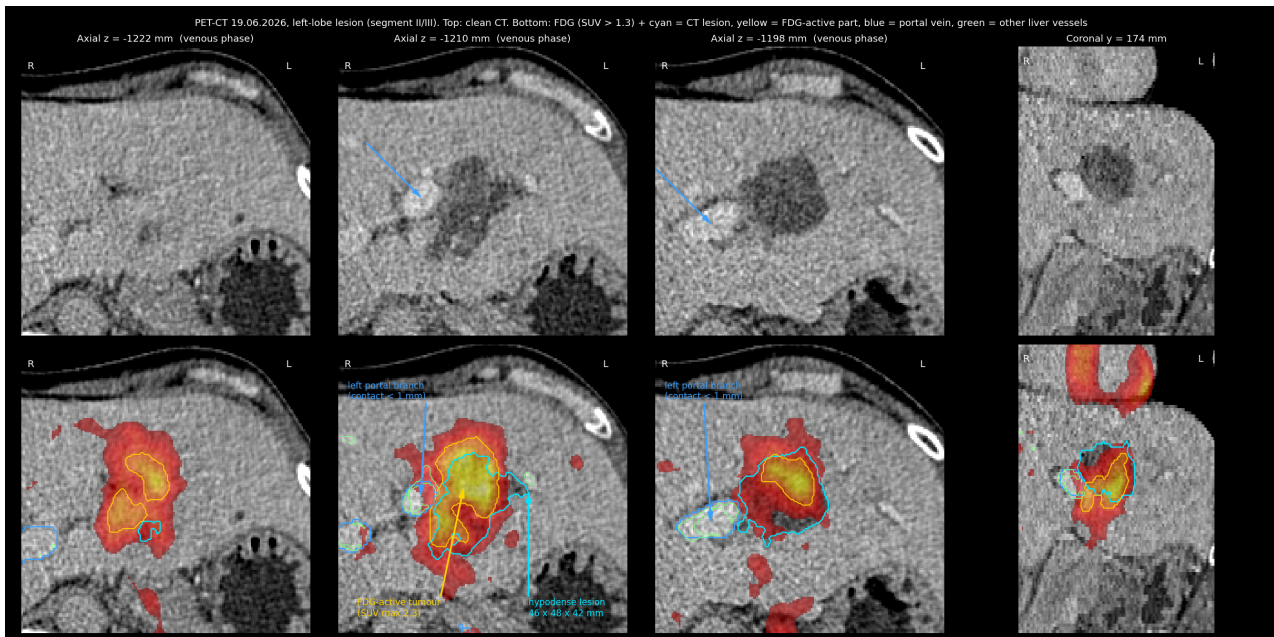
*Note on an earlier analysis.* A first automated analysis of this study (September 2026) used NIfTI volumes whose x and y axes were swapped relative to their header; laterality derived from those files was wrong and the PET was misregistered by about 7.6 mm per axis. That analysis was discarded; everything below comes from the corrected frame and agrees with the 08.09.2026 expert reading.

### 4.2 Findings

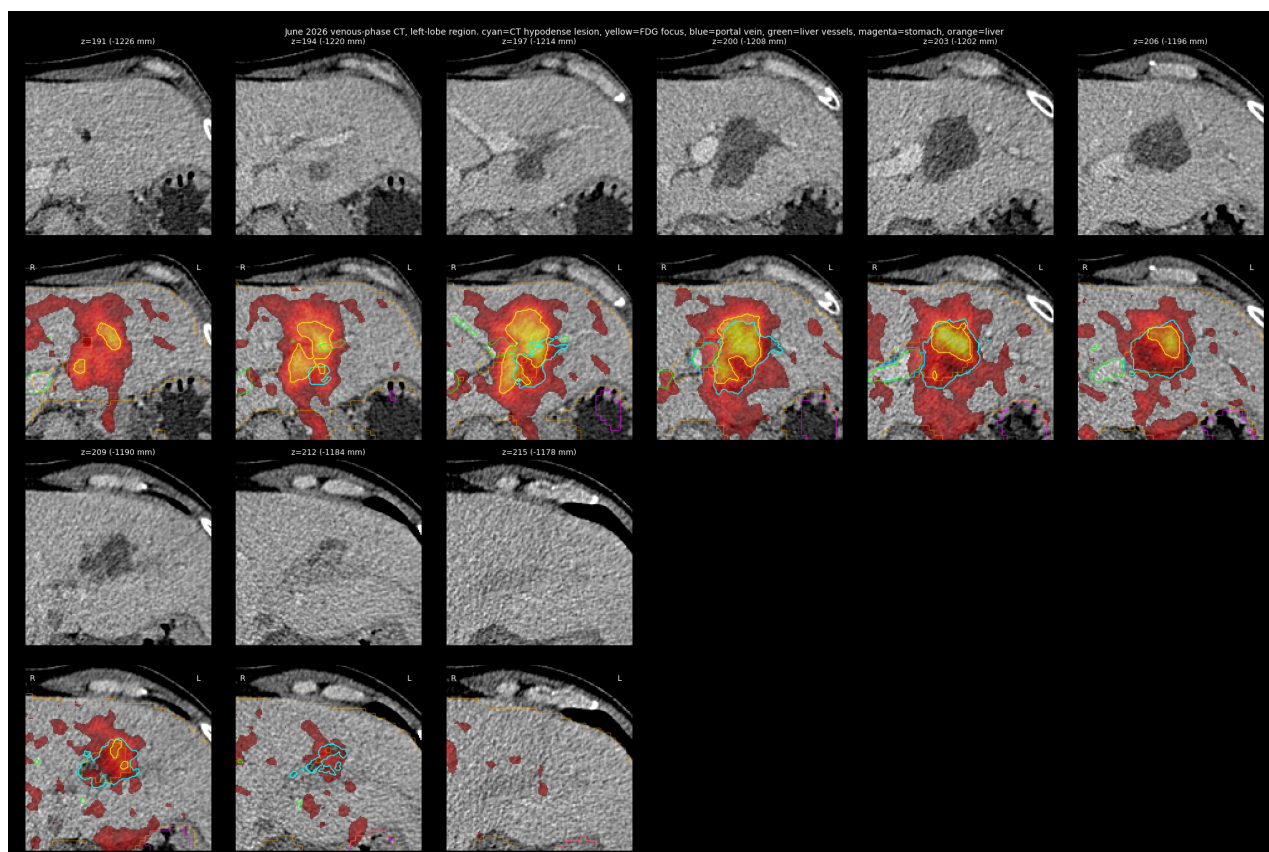
| Finding                       | Measurement (corrected frame, RAS mm)                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Left-lobe lesion, CT          | segment II (83 %) / III (17 %); hypodense, mean 44 HU vs liver 98 HU; <b>46 × 48 × 42 mm</b> , 27 mL; centre (−54, 173, −1201)                                                                    |
| FDG-active tumour             | SUVmax 2.27 (liver background 1.04); 19 mL above the 1.65 threshold; about half inside the hypodense core, the rest isodense enhancing tissue on the <b>medial (hilar) side</b> and inferior pole |
| Left portal branch            | <b>in contact</b> with the lesion (0.8 mm) and with the FDG-active part (0 mm); contact point (−35, 164, −1200)                                                                                   |
| Distances to other structures | IVC 31 mm; stomach 19 mm; heart 28 mm; the lesion reaches the liver surface (depth 0–31 mm)                                                                                                       |
| Second hottest liver focus    | SUVmax 2.34 on the anterior rim of the right-lobe mass (the “peripheral uptake” of the report)                                                                                                    |

| Finding    | Measurement (corrected frame, RAS mm)                                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bile ducts | not resolvable on CT (not dilated); the left hepatic duct runs in the same Glissonean sheath as the left portal branch, so it lies within a few millimetres of the lesion |

SUV values differ between readers and reconstructions (LISOD reported SUVmax about 3.2 for the same lesion); the relative pattern is the finding, not the absolute value. Sub-2 mm distances measured on a 2 mm CT with AI vessel masks mean “in contact”.

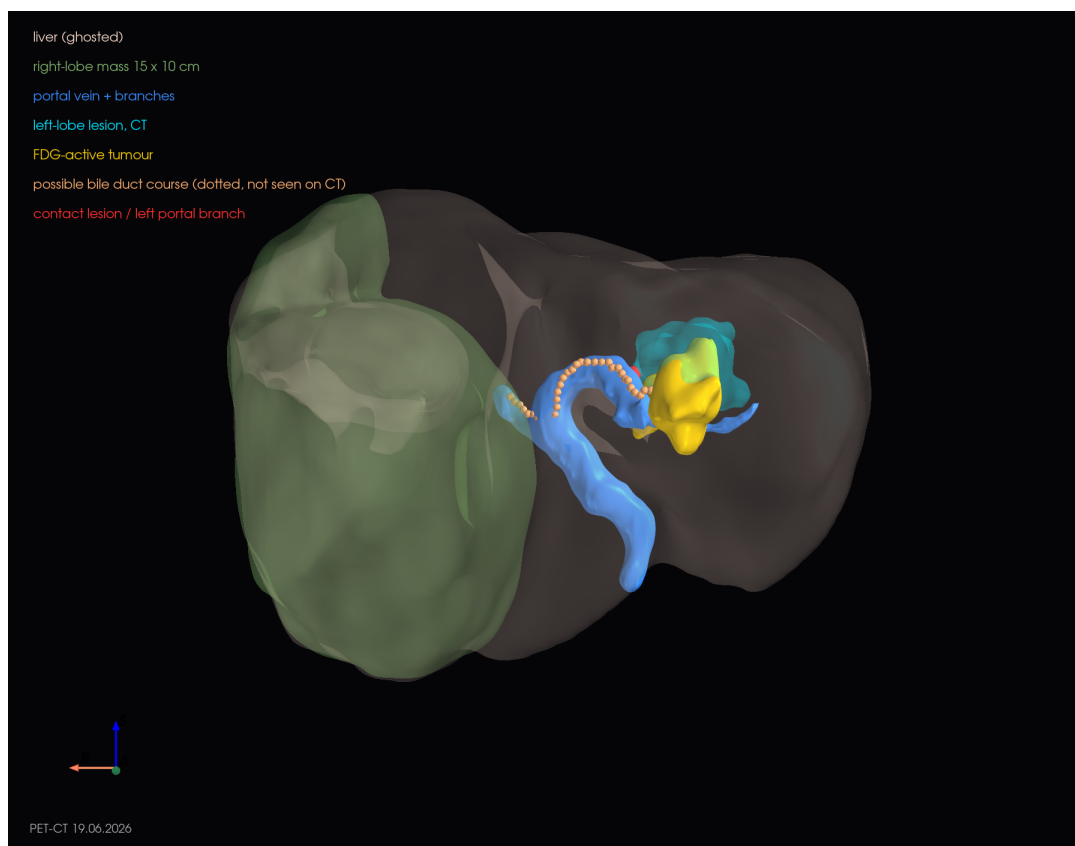


**Figure 1:** Left-lobe lesion, 19.06.2026. Top row: clean venous-phase CT (three axial levels and one coronal). Bottom row: the same slices with FDG uptake (SUV > 1.3), CT lesion outline (cyan), FDG-active tumour (yellow), portal vein (blue) and other liver vessels (green). The enhancing vessel touching the medial edge of the lesion is the left portal branch.

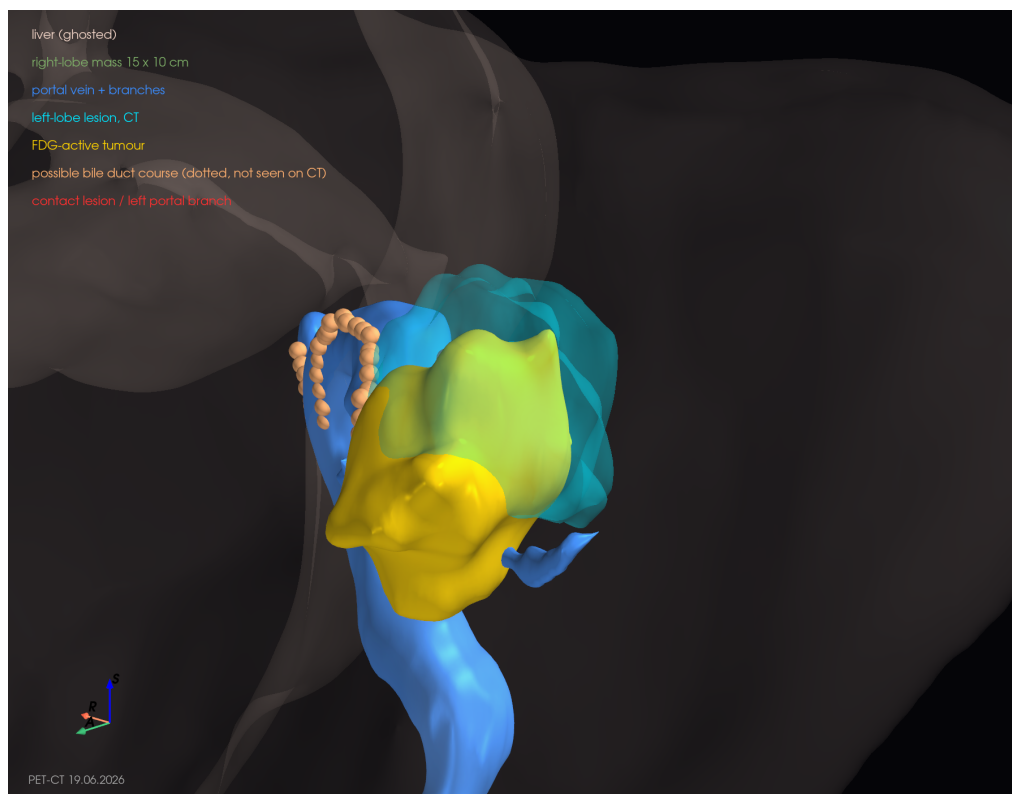


**Figure 2:** Nine consecutive axial slices (3 mm apart) through the lesion, same overlays as Figure 1.

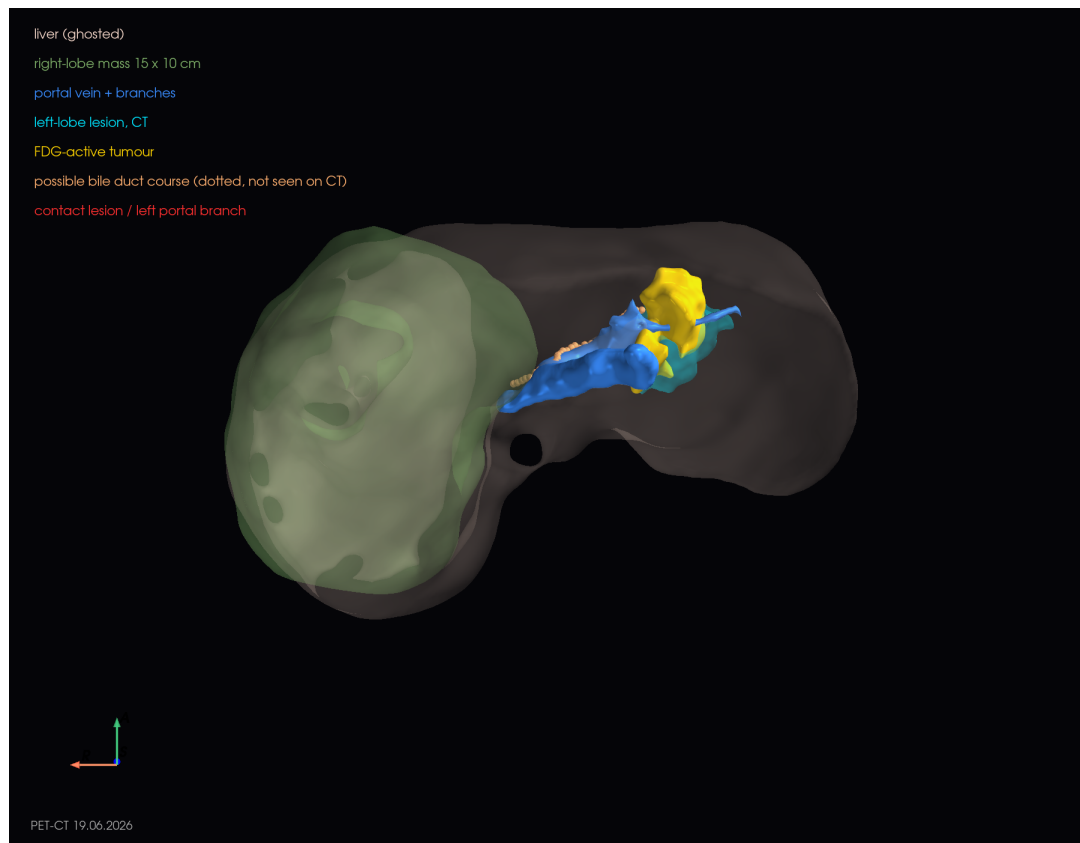
### 4.3 3D reconstruction



**Figure 3:** 3D scene, anterior view: liver ghosted, right-lobe mass translucent green, portal vein and branches blue, CT lesion translucent cyan, FDG-active tumour opaque yellow, dotted orange = possible course of the left bile duct (anatomical assumption offset antero-superior to the portal branch; not visible on CT), red = contact point.



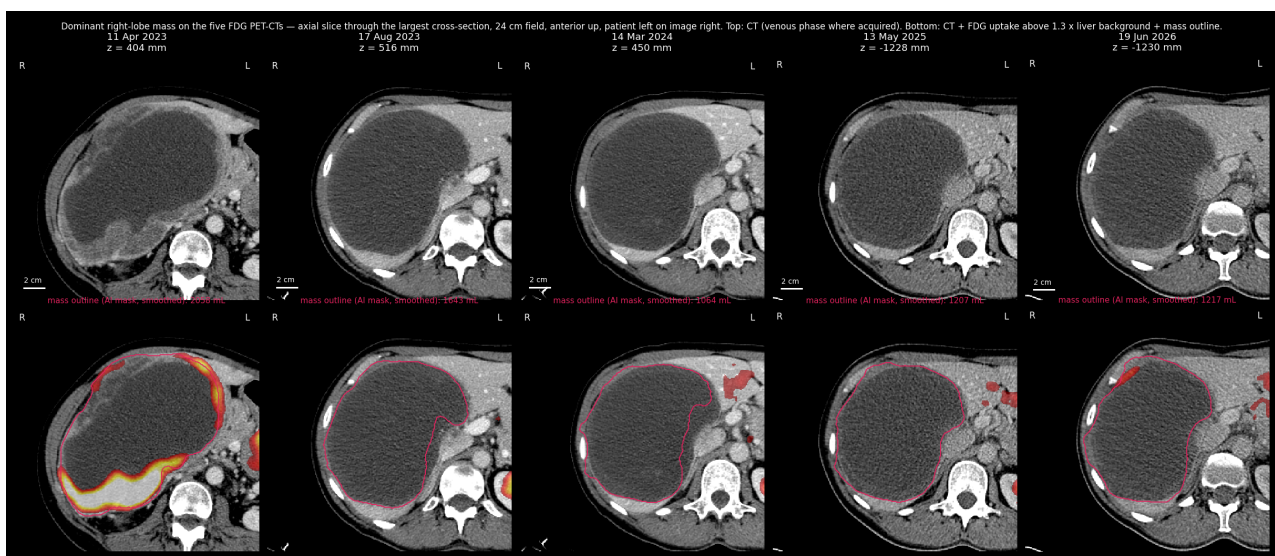
**Figure 4:** Close-up, left anterior oblique: the FDG-active tissue wraps the terminal left portal branch.



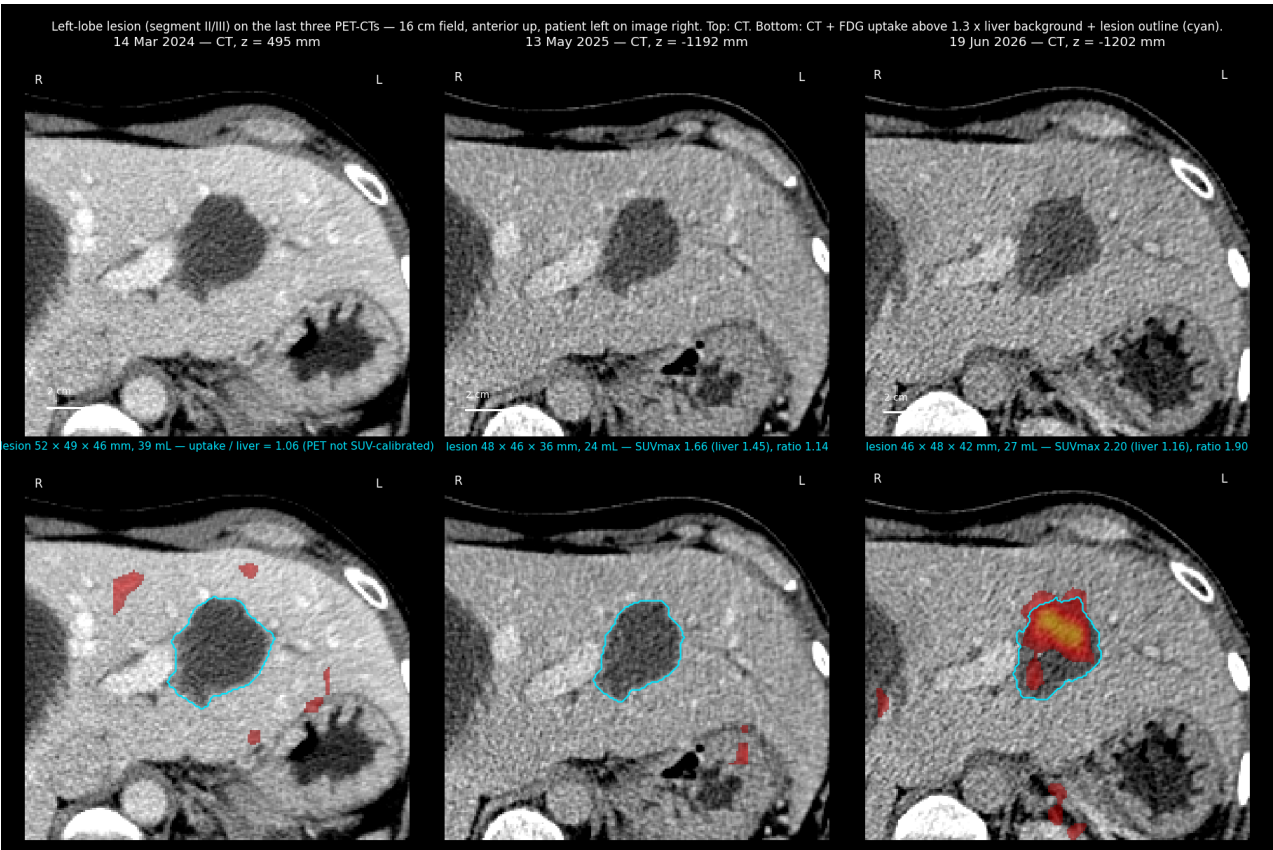
**Figure 5:** Inferior view of the same scene.

A turntable animation and an interactive glTF model are available in the Drive folder docs\_eng/2026\_new/Left\_lobe\_focus\_

## 5 Evolution of the liver disease 2023–2026



**Figure 6:** Dominant right-lobe mass through its largest cross-section on all five PET-CTs (24 cm field). Top: CT. Bottom: CT with FDG uptake above  $1.3 \times$  liver background and the automated mass outline.



**Figure 7:** Left-lobe lesion region on the last three PET-CTs (16 cm field). Top: CT. Bottom: CT with FDG uptake above 1.3 × liver background and the lesion outline.

| Study      | Left-lobe lesion, CT (mm) | Volume (mL) | SUVmax                 | Uptake / liver background |
|------------|---------------------------|-------------|------------------------|---------------------------|
| 14.03.2024 | 52 × 49 × 46              | 38.9        | PET not SUV-calibrated | 1.06                      |
| 13.05.2025 | 48 × 46 × 36              | 24.2        | 1.66 (liver 1.45)      | 1.14                      |
| 19.06.2026 | 46 × 48 × 42              | 27.4        | 2.20 (liver 1.16)      | 1.90                      |

The lesion is a pre-existing hypodense metastasis (present at the same site in March 2024 and May 2025 with uptake at or below liver background) that became FDG-active in June 2026 — the pattern of an emerging imatinib-resistant clone inside a treated lesion. Outlines were checked slice by slice on each date. The 2023–2024 PET series were not exported in SUV units, so uptake is given relative to the liver background for all dates.

**6 Gastric-wall GIST, uterus and right ovarian endometrioma 2022–2026**

On each of the ten contrast studies (five CTs of 2022–2023 from the DICOM discs and five PET-CTs) the gastric-wall mass and the pelvic mass complex were traced slice by slice from a hand-placed seed (soft-tissue density window, air and bone excluded, growth limited to tissue continuous with the seed) and every traced slice was checked visually. All five studies were acquired with intravenous contrast; the 2023–2024 PET series were not exported in SUV units, so uptake is given relative to the liver background throughout.

**6.1 Gastric-wall mass (primary GIST site)**

| PET-CT           | Max in-plane diameter (mm) | Volume (mL) | Mean HU | Max uptake / liver         | Reported                                                               |
|------------------|----------------------------|-------------|---------|----------------------------|------------------------------------------------------------------------|
| 23 Aug 2022 (CT) | 55                         | 53          | 63      | CT only                    | 51 × 46 mm                                                             |
| 1 Dec 2022 (CT)  | 53                         | 65          | 65      | CT only                    | 56 × 46 mm (German re-read 5.2 × 6.1 cm)                               |
| 4 Jan 2023 (CT)  | 52                         | 70          | 58      | CT only                    | 65 × 50 mm                                                             |
| 9 Feb 2023 (CT)  | 50                         | 52          | 51      | CT only                    | 55 × 40 mm                                                             |
| 11 Apr 2023      | 54                         | 36          | 75      | 5.5 × (not SUV-calibrated) | 58 mm, SUVmax 8.2, active                                              |
| 28 Jun 2023 (CT) | 54                         | 47          | 43      | CT only                    | 47 × 35 mm, low density                                                |
| 17 Aug 2023      | 48                         | 46          | 41      | 1.8 × (not SUV-calibrated) | 44 mm, almost fluid, inactive                                          |
| 14 Mar 2024      | 49                         | 35          | 45      | 1.3 × (not SUV-calibrated) | 40 mm, inactive                                                        |
| 13 May 2025      | 43                         | 23          | 40      | 1.8 ×                      | 35 mm, calcifications                                                  |
| 19 Jun 2026      | 41                         | 24          | 38      | 1.1 ×                      | 43 × 33 × 44 mm (Feofaniya); 37 × 31 mm, calcified, unchanged (Israel) |

The gastric-wall tumour collapsed on imatinib: from an enhancing, intensely active mass in April 2023 (about 5.5 × liver) to a larger but low-density, almost fluid, near-inactive residue by August 2023 (cystic change, as the report describes), then a plateau at about 4 cm through 2024–2026 with progressive calcification and uptake at or near liver background — in agreement with the reports. In April 2023 the mass was traced from its FDG-avid volume (restricted to soft-tissue density) because its enhancement matched the adjacent tissue on CT.

Gastric-wall GIST through its largest cross-section on all ten contrast studies 2022–2026 (12 cm field); amber titles = contrast CT, cyan titles = PET-CT with FDG above 1.3 × liver background. Cyan = traced outline.

**Figure 8:** Gastric-wall GIST through its largest cross-section on all ten contrast studies 2022–2026 (12 cm field); amber titles = contrast CT, cyan titles = PET-CT with FDG above 1.3 × liver background. Cyan = traced outline.

3D reconstruction of the traced gastric-wall mass on the ten dates at one scale (bar = 2 cm), anterior view.

**Figure 9:** 3D reconstruction of the traced gastric-wall mass on the ten dates at one scale (bar = 2 cm), anterior view.

The same masses with ghost organs from the automatic segmentation of each study, clipped to a box around the mass: stomach (lilac), pancreas (sand), spleen (rose), liver edge (pale), left kidney (grey); bar = 5 cm.

**Figure 10:** The same masses with ghost organs from the automatic segmentation of each study, clipped to a box around the mass: stomach (lilac), pancreas (sand), spleen (rose), liver edge (pale), left kidney (grey); bar = 5 cm.

## 6.2 Uterus with leiomyomas and right adnexal mass (whole pelvic complex traced)

| PET-CT           | Max in-plane diameter (mm) | Volume (mL) | Mean HU | Max uptake / liver         | Reported                                                                      |
|------------------|----------------------------|-------------|---------|----------------------------|-------------------------------------------------------------------------------|
| 23 Aug 2022 (CT) | 67                         | 175         | 59      | CT only                    | fundal mass 93 × 63 mm; right adnexal cyst 67 × 57 mm                         |
| 1 Dec 2022 (CT)  | 82                         | 167         | 53      | CT only                    | fundal mass 87 × 64 mm                                                        |
| 4 Jan 2023 (CT)  | 61                         | 145         | 44      | CT only                    | fundal mass 75 × 55 mm, uterus 45 × 42 mm; right ovarian formation 75 × 45 mm |
| 9 Feb 2023 (CT)  | 66                         | 164         | 44      | CT only                    | fundal mass 75 × 55 mm                                                        |
| 11 Apr 2023      | 96                         | 178         | 62      | 1.8 × (not SUV-calibrated) | inactive uterine mass 72 mm; right ovarian mass ~6 cm                         |
| 28 Jun 2023 (CT) | 90                         | 228         | 83      | CT only                    | pedunculated myoma 90 × 70 mm; right ovarian formation 65 × 42 mm             |
| 17 Aug 2023      | 108                        | 189         | 59      | 1.5 × (not SUV-calibrated) | 96 mm, inactive                                                               |
| 14 Mar 2024      | 98                         | 148         | 65      | 1.3 × (not SUV-calibrated) | 78 mm; Israeli revision 6.5 cm and 4.8 × 5.1 cm                               |
| 13 May 2025      | 108                        | 134         | 55      | 1.6 ×                      | 84 mm                                                                         |
| 19 Jun 2026      | 92                         | 153         | 51      | 2.4 ×                      | 68 × 55 × 48 mm + 46 × 37 mm, stable                                          |

The traced object is the whole pelvic complex (uterine body, its fibroids and the right ovarian endometrioma, traced as one object because fibroid, myometrium and the non-enhancing cyst cannot be separated reliably on every study), which is why it is larger than the single-mass diameters in the reports. Over four years it stays between 130 and 190 mL with no consistent trend and low uptake (1.3–1.8 × liver; the 2.4 × in June 2026 comes from a bowel loop touching the outline) — the behaviour of leiomyoma, not of the GIST metastases. Our numbers exceed the single-mass diameters in the reports because the whole complex is traced; on some dates the outline may include an adjacent bowel loop.

The pelvic complex through its largest cross-section on all ten contrast studies (19 cm field); amber titles = contrast CT, cyan titles = PET-CT with FDG above 1.3 × liver background. Cyan = traced outline.

**Figure 11:** The pelvic complex through its largest cross-section on all ten contrast studies (19 cm field); amber titles = contrast CT, cyan titles = PET-CT with FDG above 1.3 × liver background. Cyan = traced outline.

3D reconstruction of the traced pelvic complex on the ten dates at one scale (bar = 2 cm), anterior view.

**Figure 12:** 3D reconstruction of the traced pelvic complex on the ten dates at one scale (bar = 2 cm), anterior view.

The same masses with ghost organs: bladder (yellow), hip bones, sacrum and femoral heads (grey), colon (sand), clipped to a box around the mass; bar = 5 cm.

**Figure 13:** The same masses with ghost organs: bladder (yellow), hip bones, sacrum and femoral heads (grey), colon (sand), clipped to a box around the mass; bar = 5 cm.

## 7 Feasibility of liver-directed treatment

### 7.1 Opinion of the interventional radiologist (Dr. Oleksandr Korshak, Shalimov National Scientific Centre of Surgery and Transplantology, Kyiv; 10–11.09.2026, messenger consultation)

Thermal ablation (RFA, MWA) is contraindicated for this lesion because two contraindications coincide: proximity to the left branch of the portal vein and, above all, direct adjacency to the left bile duct. Y-90 radioembolisation (TARE) costs from about EUR 40,000 and is not performed in Ukraine (nor in Czechia or Slovakia); whether transarterial chemoembolisation (DEB-TACE/TACE) or surgical resection is possible was still under discussion. On 11.09.2026 the same specialist was asked about focused-ultrasound histotripsy (HistoSonics, newly approved in Europe): a new and very expensive method without results in large numbers of GIST patients or for lesions of this size, results from other liver tumours cannot be transferred; he suggested asking HistoSonics directly.

**Consistency with the imaging:** the image analysis supports this opinion. The lesion abuts the left portal branch and the metabolically active tumour is exactly on that side, so an ablation margin would have to include the branch and the accompanying duct.

### 7.2 Guideline basis for local treatment of focal progression

NCCN (von Mehren et al., JNCCN 2022;20(11):1204–1214; NCCN Guidelines for Patients, GIST 2023) and ESMO–EURACAN–GENTURIS (Casali et al., Ann Oncol 2022;33:20–33) both recommend, for limited progression on imatinib, continuing the TKI and treating the progressing lesion locally (resection if feasible, ablation, embolization or chemoembolization), reserving a change of systemic drug for generalised progression. In the Dutch GIST Registry (Brink et al., Eur J Surg Oncol 2023;49:106942) 123 of 457 metastatic patients had elective surgery or ablation; local treatment was associated with better overall survival (HR 0.56), with the best results when disease was confined to the liver (HR 0.27) and worse results when it was progressing on systemic treatment (HR 3.9). Ablation series for focal progression on imatinib (Dileo 2004, Hasegawa 2007, Avritscher and Gupta 2009, summarised at gistsupport.org) report complete ablation without complications and progression-free intervals of 8–16 months, with the caveat that RFA applies to lesions “not close to blood vessels or ducts” — the reason it was judged contraindicated here.

### 7.3 Evidence summary (OpenEvidence review, 05.09.2026)

- **Pattern.** Imatinib-refractory, liver-limited oligoprogression (“nodule within a mass”) driven by polyclonal secondary KIT resistance, while the exon-11 bulk disease still responds to imatinib. A combined local-plus-systemic strategy is favoured over a plain TKI switch, and imatinib should continue through any local therapy (2-year PFS after RFA 75 % with continued TKI vs 29 % without).
- **Local options for a viable nodule inside a cystic mass.** Arterially directed therapy (DEB-TACE/TACE, Y-90 TARE; NCCN 2A) deposits its effect only in perfused tissue; microwave ablation (after cyst aspiration) or cryoablation for a few well-defined nodules; **SBRT** for a perivascular or central nodule unsafe for a probe (prospective GIST series: 80 % stable target lesions, median 16 months); **IRE** as the vessel- and duct-sparing alternative for a nodule abutting major vessels or central ducts (GIST data anecdotal); RFA is the least suitable in fluid-containing targets.
- **Timing.** Ablation planning needs multiphase CT or MRI no older than about 30 days; the June scan is stale and the 32 % ctDNA fraction argues for re-imaging now and definitive local therapy within about a month, provided the disease is still oligoprogressive.
- **Systemic.** Sunitinib covers V654A but not A829P; regorafenib and ripretinib cover A829P and regorafenib remains clinically reasonable against V654A (VOYAGER ctDNA data); sorafenib has the broadest in-vitro coverage but only retrospective GIST evidence. The PRES history raises the stakes of potent VEGFR inhibitors: reduced starting dose and strict blood-pressure monitoring. Clinical-trial referral is recommended.

- **Extrahepatic disease.** The femoral-head lesion flagged by LISOD should be characterised (MRI), because extrahepatic disease reduces the value of arterial therapies; the Israeli reading found no bone lesions.

## 7.4 Proposed systemic drugs and coverage of the three KIT clones

Merged from the [FoundationOne report](#) (therapy list, NCCN categories, resistance flags) and the [OpenEvidence review](#) (subgroup data from INTRIGUE and VOYAGER, toxicity). Median PFS values are trial subgroup figures.

| Drug / approach                           | FoundationOne                       | V654A (exon 13)                                  | A829P (exon 18)              | VEGFR / PRES risk                                        | Status for her                                                         |
|-------------------------------------------|-------------------------------------|--------------------------------------------------|------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|
| Imatinib 400 mg (current)                 | resistance flagged                  | resistant                                        | resistant                    | none                                                     | continuing; keep through any local therapy                             |
| Sunitinib 50 mg 4/2                       | resistance flagged (A829P)          | <b>strong</b> (mPFS 15.0 mo)                     | resistant (2.3 mo)           | <b>high</b> (grade $\geq 3$ HTN 22.6 %, PRES warning)    | prescribed 30.07 and 08.09.2026, <b>not started</b>                    |
| Regorafenib 160 mg d1–21                  | <b>NCCN category 1</b>              | acceptable clinically (7.4 mo, VOYAGER)          | active (6.7 mo)              | <b>high</b> (HTN 10–24 %, PRES in label, hepatotoxicity) | named as option 08.09.2026                                             |
| Ripretinib 150 mg                         | limited resistance evidence         | reduced (4.0 mo)                                 | <b>strong</b> (14.2 mo)      | <b>low</b> (HTN 5.8 %, no PRES in label)                 | fourth line; equal to sunitinib in the co-occurring subgroup (HR 1.07) |
| Sorafenib                                 | NCCN 2A, other tumour type          | active in vitro (conflicting)                    | active                       | moderate                                                 | retrospective data                                                     |
| Avapritinib                               | resistance flagged                  | resistant                                        | active                       | bleeding warning                                         | not suitable                                                           |
| Nilotinib / ponatinib / dasatinib         | NCCN 2A other tumour type / no flag | no clinical activity / escape mutation / no data | —                            | low / arterial risk / low                                | not recommended                                                        |
| Velzatinib (IDRX-42), StrateGIST-1        | —                                   | active (exon 13 ORR 17 %)                        | active (exon 17/18 ORR 23 %) | selective KIT, low VEGFR                                 | trial only; 2nd line mPFS 13.7 mo, 3rd line 9.2 mo                     |
| Bezuclastinib + sunitinib (Peak, phase 3) | —                                   | yes (sunitinib)                                  | yes (bezuclastinib)          | sunitinib risk retained                                  | trial only                                                             |
| Local therapy + imatinib                  | —                                   | ablates clone                                    | ablates clone                | <b>none</b>                                              | NCCN-endorsed for limited progression; fresh imaging first             |

## 8 Questions for the multidisciplinary tumour board

1. **Physical removal of the new focus, with imatinib continued.** Thermal ablation has been declined because the 42 × 36 mm segment II/III lesion abuts the left portal branch and the left hepatic duct.

Which option does the board recommend and where can it be done: resection of the left lateral section (segments II–III), irreversible electroporation (NanoKnife), histotripsy (HistoSonics, newly approved in Europe), SBRT, or transarterial treatment (DEB-TACE/TARE) of the segment II/III feeding artery — and how should the active rim of the right-lobe mass be handled?

2. **New systemic drug or trial, if and when needed.** Both resistance clones (KIT exon 13 V654A and exon 18 A829P) must be covered, each later TKI line gives a shorter time to the next progression with more toxicity, ripretinib costs about USD 45,000 a month, and she had PRES with intracerebral haemorrhage on a VEGFR inhibitor. Which drug or trial (sunitinib as prescribed, regorafenib, ripretinib, velzatinib/IDRX-42, bezucastinib + sunitinib) would the board choose, at what point — now, or only after local control fails — and with what blood-pressure safeguards?

## 9 Appendix A — Expert opinion on the FDG PET-CT, 08.09.2026 (English translation)

Drive: [this translation](#) (Russian original)

### 9.0.1 Expert opinion on FDG PET-CT — English translation

Source (Russian original): docs\_eng/2026\_new/Zherlitsyna Anastasia - PET CT revision - 08.09.2026.docx (second medical opinion, Israel). Translation made 10.09.2026; wording follows the original as closely as possible.

#### 9.0.1.1 Patient and study details Patient: ZHERLITZINA ANASTASIA / Zherlitsyna Anastasia

ID: ME923960

Date of birth: 25.04.1976

Study date: 19.06.2026

Date of reading: 08.09.2026

#### 9.0.1.2 Indication The patient is under follow-up and receiving treatment for metastatic gastrointestinal stromal tumour (GIST).

Comparison: with the previous study of 13.05.2025.

#### 9.0.1.3 Technique The study was performed after intravenous injection of the radiopharmaceutical, without intravenous and without oral contrast agent.

#### 9.0.1.4 Findings

##### 9.0.1.4.1 Head and neck As is known, the diagnostic capability of this study with respect to the brain is limited. Within the limits of the method, no gross changes of the brain parenchyma are identified.

No hypermetabolic enlarged cervical lymph nodes are identified.

The thyroid gland is of normal size, without pathological FDG uptake.

##### 9.0.1.4.2 Chest Small nodules are seen in both lungs, stable over a long period, with sizes below the resolution of PET.

No pleural effusion.

Minimal amount of pericardial fluid — unchanged.

No pathological FDG uptake in the mediastinal and hilar lymph nodes.

No pathological FDG uptake in the breast tissue.

No pathological uptake in the axillary lymph nodes.

The superior vena cava (SVC) is not visualised — probably occluded. Dilated collateral vessels are seen in the chest.

##### 9.0.1.4.3 Abdomen and pelvis The known hypodense lesions are again identified in both lobes of the liver.

A large lesion in the right lobe of the liver with peripheral FDG uptake, which in the present study is somewhat more pronounced. The size of the lesion is about 15 × 10 cm; on the previous study — about 11–12.5 cm.

The lesion in the left lobe of the liver, about 42 × 36 mm — without significant change in size, however the intensity of FDG uptake has now increased.

The gallbladder is not enlarged. The intrahepatic bile ducts are not dilated. The portal vein is patent.

The known lesion in the stomach wall with calcifications, without significant change in size — about 37 × 31 mm. Minimal FDG uptake is noted, similar to that observed previously, without significant dynamics.

The spleen is of normal size, without pathological FDG uptake.

No pathological FDG uptake is identified in the pancreatic parenchyma.

The right adrenal gland is compressed by the liver, without mass lesions and without pathological FDG uptake.

Thickening of the left adrenal gland without pathological FDG uptake — unchanged.

Kidneys without radiopaque calculi, no hydronephrosis.

No hypermetabolic enlarged lymph nodes are identified in the abdominal cavity, retroperitoneum or pelvis.

No pathological FDG uptake in the inguinal lymph nodes.

Urinary bladder without significant changes.

Bowel loops, as far as can be assessed by this method, without significant changes.

The known heterogeneous lesions of the uterus — without significant dynamics.

**9.0.1.4.4 Skeletal system** Within the examined area no pathological foci of FDG uptake or destructive bone changes are identified.

**9.0.1.5 Conclusion** The known gastric lesion — without significant change in size and without significant FDG uptake.

Metabolic progression of the liver lesions, especially of the lesion in the left lobe, as well as enlargement of the large lesion in the right lobe of the liver, whose metabolic activity has also somewhat increased.

Small pulmonary nodules — unchanged.

**9.0.1.6 Signature** Dr. Victoria Kulikova

Nuclear medicine specialist

Physician licence No. 84847

Specialist licence No. 131252

Translator's note: the report states the CT was performed without intravenous contrast; the DICOM series of the 19.06.2026 study include a venous-phase contrast series (series 9), so this statement appears to refer to the PET reading protocol rather than to the CT acquisition.

## 10 Appendix B — LISOD retrospective radiology report, 08.07.2026 (English translation)

Drive: [this translation](#) (Ukrainian original)

### 10.0.1 LISOD Radiology Report — Retrospective PET-CT Analysis (English Translation)

Source (Ukrainian original): docs\_eng/anastasia\_disks/June\_19\_2026\_PET\_CT/Жерліцина\_Анастасія\_Станіславівна\_F

Facility: LISOD, Radiology Diagnostics Department

Patient: Zherlitsyna Anastasiia Stanislavivna, ID 110889, DOB 25.04.1976

Report date: 08.07.2026

Study: retrospective analysis of provided imaging studies (PET/CT with FDG from 19.06.2026), compared to prior PET/CT with FDG from 13.05.2025 and 14.03.2024

Reporting physician: Novikov M.Ye.

**10.0.1.1 Clinical background (as stated in report)** ICD-10: C16.9 - Malignant neoplasm of stomach, unspecified. Diagnosis: Leiomyosarcoma of uterine body, T2NxM1, secondary liver and peritoneal involvement. Status after 4 cycles of Gemcitabine + Docetaxel + Avastin chemotherapy with liver progression (per CT/MRI 10/2022). C-KIT, MET mutation detected (ONCOMINE). Status after Pazopanib treatment (05-12/01/2023) with PRES development and positive CT dynamics. 01/2023 - repeat liver biopsy, histology: uterine leiomyosarcoma metastasis. Status after 3 cycles of Trabectedin with negative radiological dynamics (PET/CT 11/04/2023). Suspected synchronous gastric neoplasia (GIST?). Status after EUS-FNA gastric biopsy, liver core biopsy - histologically GIST of stomach, GIST metastasis to liver (CSD). 08/2023 - status after 4 months Imatinib with pronounced positive PET/CT response. 03/2025 - status after 2 years Imatinib, positive clinical picture. 05/2025 - stable radiological dynamics on Imatinib.

**10.0.1.2 Findings** Head and neck: No changes characteristic of metabolically active aggressive neoplastic involvement, same as prior imaging. No significant change since 13/05/2025, no new active or suspicious lesions.

Chest: No changes characteristic of metabolically active aggressive neoplastic involvement, same as prior imaging. No significant change since 13/05/2025, no new active or suspicious lesions. No excess pleural/pericardial fluid, no large vessel thrombi.

Abdomen and pelvis: Known foci and masses in the liver are generally stable in size overall; however, mural (wall) components within them — particularly in the largest lesion in the right lobe — show new nodular elements with notable enhancement and partial focal moderately increased activity, moderately higher than the hepatic blood pool. This is best seen in nodule "42" in the left lobe, with uptake up to SUVmax 3.2 (adjacent parenchyma SUV 2).

No new separate similar active or suspicious nodules found.

Known residual gastric tumor without significant activity; nodule above the bladder/uterus on the right also without significant activity and without change.

No new separate active or similar nodules, no new suspicious lymphadenopathy found.

No signs of bowel obstruction, ascites, hydronephrosis, bile duct dilation, or large vessel thrombi.

Bones and soft tissue: New elongated/geographic lucency in the femoral head of the right hip, approximately 31mm, with a sclerotic rim and moderate activity, currently without flattening/deformity of the head. No other new active/suspicious bone lesions found.

**10.0.1.3 Radiological conclusion** New mural (wall) components with moderate activity within the liver lesions, raising suspicion of return of viable tumor elements of GIST metastases.

Physician: Novikov M.Ye.

[Translation note: plain-language English translation of the Ukrainian original for reference purposes; not a certified medical translation.]

## 11 Appendix C — PET-CT report, Feofaniya, 19.06.2026 (English translation)

Drive: [this translation](#) (Ukrainian original) · DICOM disc

### 11.0.1 PET-CT Whole Body with Contrast Enhancement — English Translation

Source (Ukrainian original): docs\_eng/anastasia\_disks/June\_19\_2026\_PET\_CT/Жерліцина\_Анастасія\_Станіславівна\_f4

Facility: All-Ukrainian Radiosurgery Center (Feofaniya), PET-CT Diagnostics Department

Patient: Zherlitsyna Anastasiia Stanislavivna, b. 1976

Study: Whole-body PET-CT with contrast enhancement

Date: 2026-06-19 / #53478/25

F18-FDG injected IV: 232.7 MBq; PET effective dose 4.4 mSv, CT 18.7 mSv

**11.0.1.1 Clinical data** 1. Gastrointestinal stromal tumor (GIST) of the stomach, GIST metastasis to the liver (CSD). Status post EUS-FNA biopsy of the stomach, core biopsy of the liver. 08.2023 - status after 4 months of Imatinib treatment. 03.2025 - status after 2 years of Imatinib treatment with positive clinical picture. Daily Imatinib (Glivec) continued after the last PET-CT.

2. Leiomyosarcoma of the uterine body T2NxM1, secondary involvement of liver and peritoneum. Status after 4 courses of chemotherapy per Gemcitabine+Docetaxel+Avastin protocol with disease progression in the liver (10.2022). C-KIT and MET mutations detected. Status after Pazopanib treatment (05-12.01.2023) with development of posterior reversible encephalopathy syndrome and positive CT dynamics. 01.2023 repeat liver biopsy: histology - metastasis of uterine leiomyosarcoma. Status after 3 cycles of chemotherapy per Trabectedin protocol with negative dynamics (PET-CT from 11.04.2023).

Reference PET-CT for comparison: 13.05.2025

Reference values: mediastinal blood pool SUVmax 1.41; liver parenchyma pool SUVmax 1.99.

### 11.0.1.2 Findings Oncological disease control:

Stomach: not distended; along the lesser curvature a hypodense ametabolic mass persists, stable in size at up to 4.3×3.3×4.4 cm (ax+sag), with increasing calcifications in its structure.

Distant metastatic spread:

Liver: compensatorily enlarged, right lobe craniocaudal size up to 18.7 cm (13.05.2025 - up to 17.9 cm).

Parenchyma contains ametabolic, hypodense masses, stable in size:

- Sg5/6/7/8: massive mass up to 14.4×10.4×14.1 cm (ax+cor)
- Sg7: satellite lesion up to 1.0×1.0 cm
- Sg7/8: subcapsular, up to 6.3×4.3 cm
- Sg5/8 border: small lesion up to 0.6×0.6 cm
- Sg4a/7 border: up to 0.6×0.6 cm
- Sg2/3 border: mass along the left portal vein, 4.5×3.6 cm
- Sg1: up to 1.3×1.3 cm
- Sg4b: small lesion up to 0.4×0.4 cm

Uterus: along the anterior wall, a heterogeneous round mass with calcifications, conditionally stable in size at up to 6.8×5.5×4.8 cm (ax+sag) - per history, likely treated leiomyosarcoma. Along the upper margin, an ametabolic mass of stable size up to 4.6×3.7 cm - pelvic MRI with contrast recommended for correct assessment. Ovaries not clearly differentiated - correlate with ultrasound data.

Bones unremarkable, no suspicious bone marrow FDG uptake.

Neck: physiological symmetric FDG metabolism in the palatine tonsils. No signs of metabolically active cervical lymphadenopathy. Thyroid not enlarged, normal structure, no increased FDG uptake. A subcutaneous port system is placed at the right subclavian region; catheter passes into the superior vena cava, distal tip at the lower third of the SVC.

Chest: mammary glands without pathological FDG uptake. No signs of metabolically active axillary, mediastinal or hilar lymphadenopathy. Trachea and main bronchi patent. Lung parenchyma normal aeration, no infiltrative or mass changes. Single small bullae on the right in S1, subpleural, up to 0.8 cm. No pleural effusion.

Abdomen/retroperitoneum: gallbladder not enlarged, compressed by adjacent liver parenchyma, contents of homogeneous density, no radiopaque calculi. Pancreas normotrophic, contour clear and scalloped. Adrenal glands not enlarged, normal shape. Kidneys typically positioned, parenchyma normal thickness. Collecting systems and ureters normal configuration. No metabolically active intra-abdominal or retroperitoneal lymphadenopathy.

Pelvis: bladder moderately filled, walls not thickened, no additional masses. Small amount of fluid in the Pouch of Douglas. No metabolically active pelvic or inguinal lymphadenopathy.

**11.0.1.3 Conclusion** Compared with the PET-CT from 13.05.2025 (note: report references this as 13.05.2026, likely a typographical date error for 13.05.2025), the picture of the known gastrointestinal stromal tumor (GIST) of the stomach and metastatic liver involvement shows no reliable negative dynamics.

Given the histologically confirmed GIST, which may show absent or variable FDG avidity, the absence of pronounced metabolic activity does not exclude residual tumor tissue. For correct assessment of disease course and further monitoring, correlation with contrast-enhanced CT or MRI is recommended.

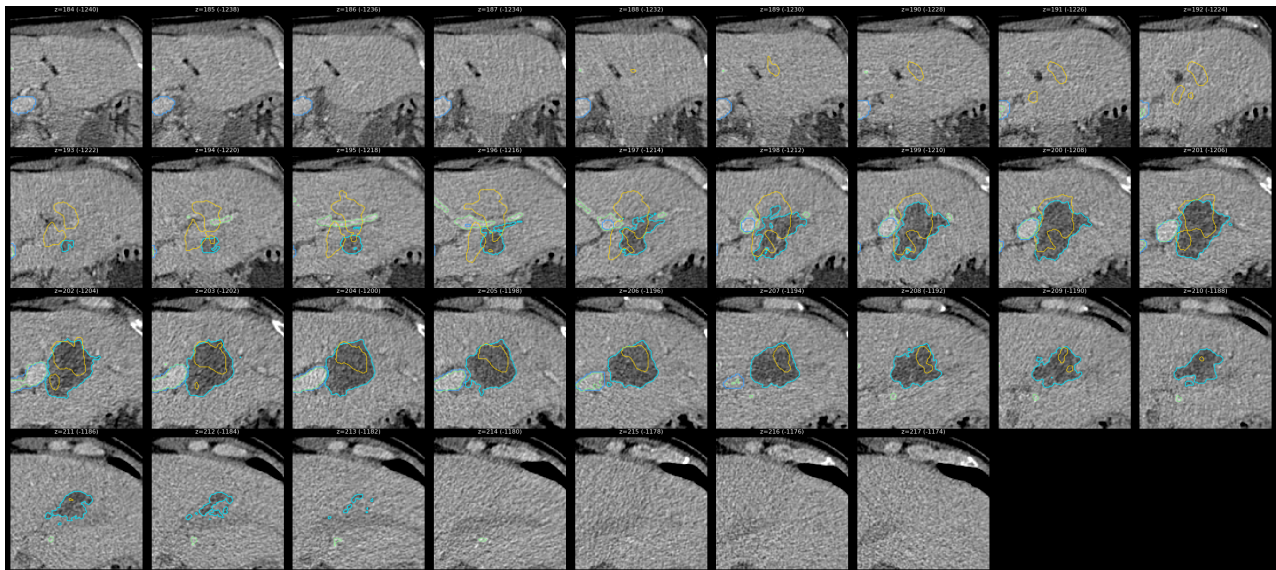
The calcified uterine body mass is stable over time; the ametabolic mass along the upper margin of the uterine body - pelvic MRI with contrast is advisable for correct assessment.

Radiologist: Zakrzhevskya K.P.

Head of the All-Ukrainian Radiosurgery Center, radiologist: Kmetiuk Ya.V.

[Translation note: this is a plain-language English translation of the Ukrainian original for reference purposes; it is not a certified medical translation.]

## 12 Appendix D — Slice-by-slice verification of the lesion outline



**Figure 14:** Every 2 mm slice through the lesion: cyan = CT lesion outline, yellow = FDG-active tumour, blue = portal vein, green = other vessels.